

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Scala for D.S Practical

Practical - 25,26

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Class :SY-Bsc.IT	Batch : 00
Date of Assignment :-01-09-2026	Date/Time of Submission : 01-09-2026

PRACTICAL 25

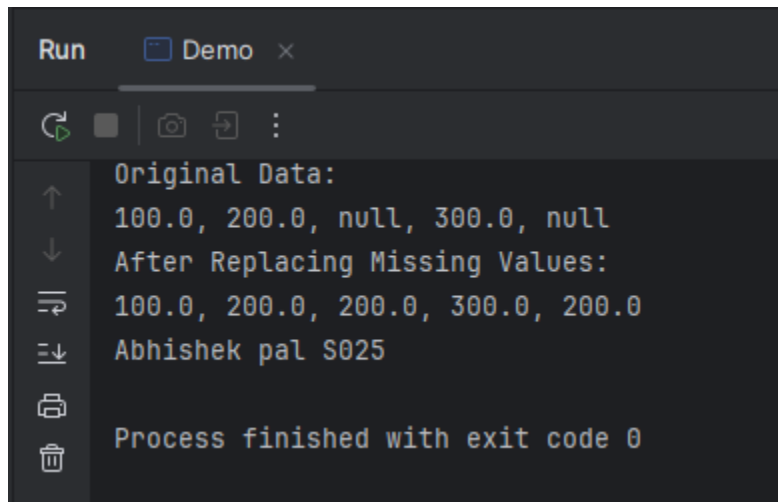
Aim:

Handle missing values in a dataset. Replace missing values with the column mean.

Code:

```
build.sbt  Demo25.scala x
1  object Demo extends App {
2
3      val sales: List[java.lang.Double] = List(100.0, 200.0, null, 300.0, null)
4
5      val v = sales.filter(_ != null)
6
7      val mean = v.map(_.toDouble).sum / v.length
8      println("Mean Sales: " + mean)
9
10     val n = sales.map { value =>
11         if (value == null) mean
12         else value
13     }
14     println("Original Data:")
15     println(sales.mkString(", "))
16     println("After Replacing Missing Values:")
17     println(n.mkString(", "))
18
19     println("Divyanshu Singh S039")
20 }
21
Demo
```

Output:



The screenshot shows a terminal window titled "Run" with a sub-tab "Demo". The terminal output is as follows:

```
Original Data:
100.0, 200.0, null, 300.0, null
After Replacing Missing Values:
100.0, 200.0, 200.0, 300.0, 200.0
Abhishek pal S025
Process finished with exit code 0
```

The terminal interface includes a dark-themed toolbar with icons for running, pausing, copying, pasting, and other standard terminal functions.

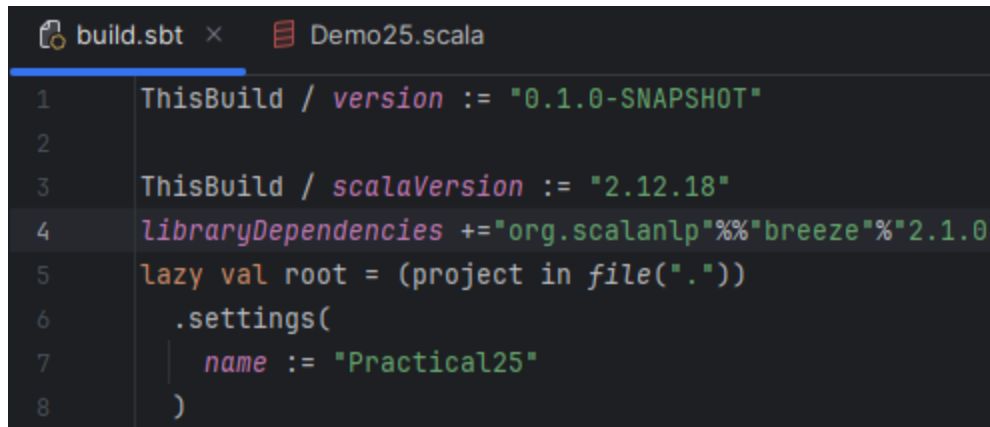
PRACTICAL 26

Aim:

Perform logistic regression using Breeze. Classify a dataset with binary labels.

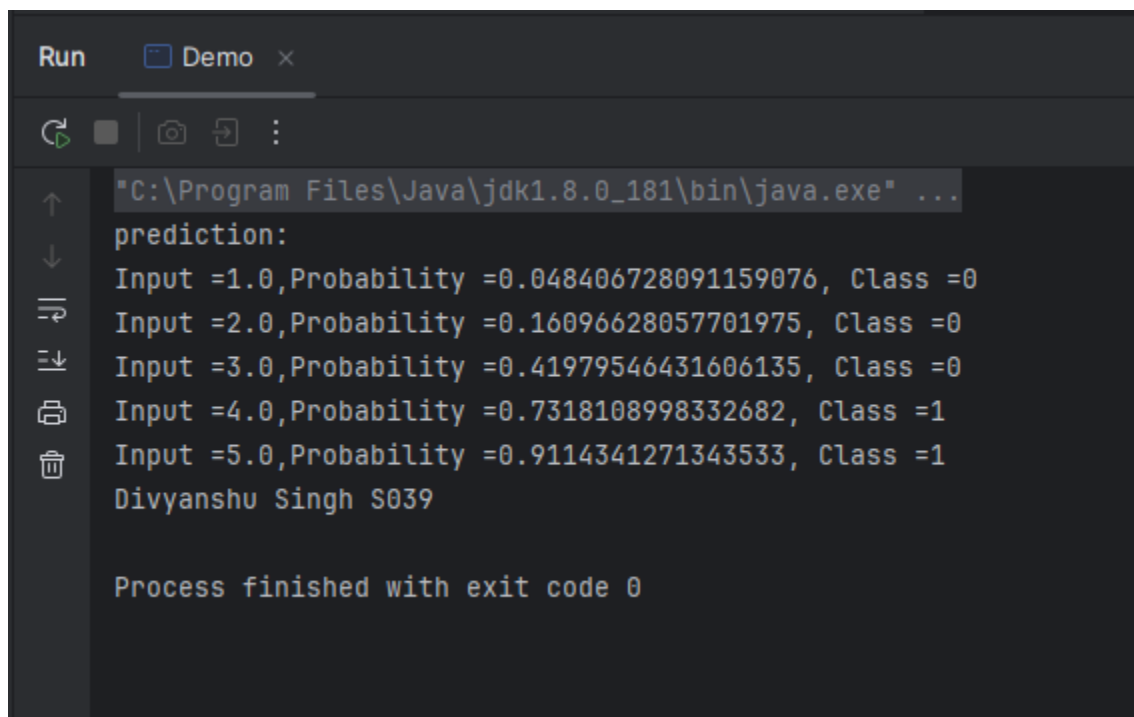
Code:

```
build.sbt  Demo25.scala x
1
2 import breeze.linalg._
3 object Demo extends App {
4
5     val x = DenseVector(1.0,2.0,3.0,4.0,5.0)
6     val y = DenseVector(0.0,0.0,0.0,1.0,1.0)
7
8     var weight = 0.0
9     var bias = 0.0
10
11     def sigmoid(z: Double): Double = {
12         1.0/(1.0+ math.exp(-z))
13     }
14     for (i <- 1 to 1000){
15         for (j <- 0 until x.length){
16             val prediction = sigmoid( weight *x(j)+bias)
17             val error = prediction -y(j)
18             weight = weight -0.02*error*x(j)
19             bias = bias -0.01*error
20         }
21     }
22     println("prediction:")
23     for(i<-0 until x.length){
24         val probability = sigmoid(weight * x(i)+bias)
25         val result =
26             if (probability >= 0.5)1 else 0
27         println("Input =" +x(i)+",Probability =" + probability +", Class =" +result)
28     }
29     println("Divyanshu Singh S039")
30 }
```



```
build.sbt x Demo25.scala
1 ThisBuild / version := "0.1.0-SNAPSHOT"
2
3 ThisBuild / scalaVersion := "2.12.18"
4 libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
5 lazy val root = (project in file("."))
6   .settings(
7     name := "Practical25"
8   )
```

Output:



```
Run Demo x
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
prediction:
Input =1.0,Probability =0.048406728091159076, Class =0
Input =2.0,Probability =0.16096628057701975, Class =0
Input =3.0,Probability =0.41979546431606135, Class =0
Input =4.0,Probability =0.7318108998332682, Class =1
Input =5.0,Probability =0.9114341271343533, Class =1
Divyanshu Singh S039

Process finished with exit code 0
```